
title: EcoNudge SDK
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EcoNudge SDK: Project Documentation

1. Problem Overview

Global climate change is accelerating, driven heavily by consumer micro-decisions. While many consumers possess a desire to reduce their carbon footprint, **friction kills adoption**.

Currently, living sustainably requires users to download separate carbon-tracking applications, manually log their daily activities, and attempt to offset their emissions post-facto. This creates massive cognitive load. In the e-commerce, food delivery, and ride-sharing sectors, carbon emissions are hidden behind the convenience of a "Book Now" button.

The core problem: We don't need another standalone eco-app. We need an invisible sustainability layer integrated directly into the platforms where users already make high-carbon decisions.

2. Solution Approach & System Design

EcoNudge SDK solves this by intercepting decisions at the point of action. We built a drop-in React SDK that enterprise platforms (e.g., Uber, Zomato) can embed into their checkout flow. When a user clicks "Book", our proprietary engine calculates the carbon footprint in under 50ms and injects a real-time, gamified UI suggesting a greener alternative (e.g., "Take an auto instead. Save 192g of CO₂").

Architecture Diagram

Below is the system architecture demonstrating how the EcoNudge ecosystem operates seamlessly with sub-50ms latency:

```

graph TD
  %% User Flow
  User([End User]) -->|Clicks Checkout| HostApp[Partner App: Uber/Amazon]

  %% SDK Integration
  HostApp -->|Init Checkout| SDK[EcoNudge React SDK]

  subgraph EcoNudge Edge Engine
    SDK -->|Intercept Request| Engine[Carbon Predictive Engine]
    Engine <-->|Query <50ms| DB[(Emission Factors DB)]
    Engine -->|Return Alternatives| NudgeUI[Dynamic Nudge Modal]
  end

  %% Decision & Reward
  NudgeUI -->|User Accepts Green Option| Gamification[Gamification Service]
  Gamification -->|Update Health Ratio| Tree[Virtual Tree & City Wars]
  Gamification -->|Award| EcoPoints[EcoPoints Ledger]

  %% Enterprise Loop
  HostApp -->|B2B Reporting| Admin[ESG Corporate Admin Portal]
  Admin -->|Verify| ESG[Global ESG Compliance]

  %% Monetization / Real World
  EcoPoints -->|Redeem| Market[Offset Marketplace]
  Market -->|Fund| RealWorld[Real-World Tree Planting / Solar]

```

3. Key Features

A. Sub-50ms Predictive Carbon Engine

To ensure we do not harm our partner platforms' checkout conversion rates, our algorithm processes over 1,000+ data points (distance, packaging, vehicle type) in real-time. This Edge-based computation ensures the widget loads instantly without blocking the main UI thread.

B. Gamification & Retention Loop

We transformed carbon tracking into a multiplayer game:

- **The Virtual Tree:** A Tamagotchi-style WebGL asset that thrives or withers based on your physical actions.
- **City Wars:** A global, low-latency leaderboard where cities compete in real-time to save the most carbon.

C. Generative AI Reporting

Instead of static numbers, we built a personalized LLM hook. The on-board AI analyzes the user's specific micro-habits and types out a long-form, dynamic sustainability report directly into the dashboard.

D. Corporate ESG Admin Portal

For enterprise partners, integrating EcoNudge provides verifiable, real-time Environmental, Social, and Governance (ESG) data. This allows corporations to instantly report their customer-driven carbon offsets to global regulators, making EcoNudge incredibly profitable for B2B integration.

E. Developer API & Offset Marketplace

We give developers full headless API access to our carbon-calculation engine. Furthermore, users can spend their earned EcoPoints in our Offset Marketplace to fund actual, physical planetary healing, closing the digital-to-physical loop.